



Patuakhali Science and Technology University

Dumki, Patuakhali

Assignment

Course code: CCE-421

Course title: Cryptography and Network Security

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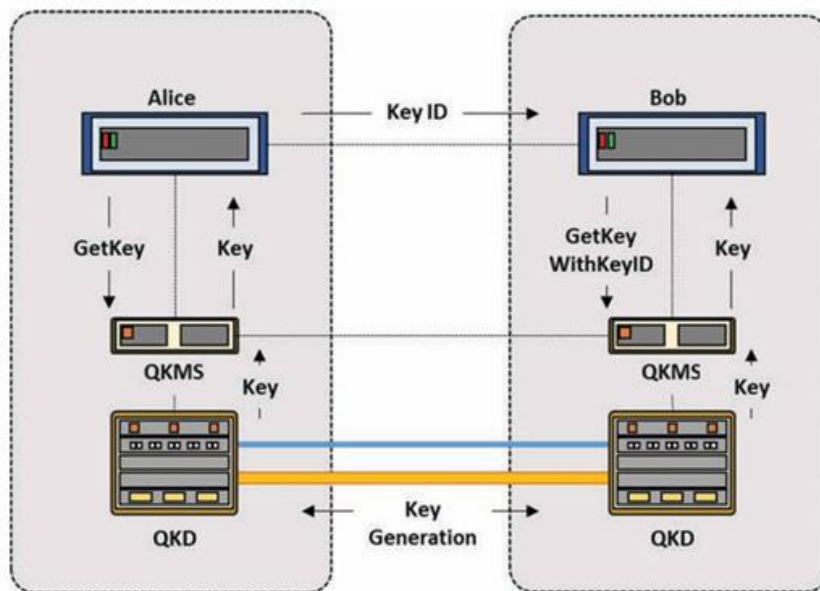
1. How does QKD provide security, and why can it detect eavesdropping? (7 marks)

Answer:

Quantum Key Distribution (QKD) provides security by allowing two users (often called *Alice* and *Bob*) to create a shared secret key using quantum states (e.g., photons), instead of sending the key directly over the network.

Security idea (key sharing without exposing the key)

- In QKD, Alice sends quantum bits (qubits) to Bob through a quantum channel.
- After transmission, they use a public classical channel to compare some settings (not the full key) and keep only the correct bits.
- The final result is a shared random key that can later be used for normal encryption (like AES/one-time pad).



Why it can detect eavesdropping

QKD can detect eavesdropping because of a basic quantum rule:

- Measurement disturbs quantum states.
- If an eavesdropper (*Eve*) tries to intercept and measure the qubits, she will unavoidably change some of them.

- This creates extra errors in the key bits that Alice and Bob can measure by publicly comparing a small sample of their bits (error-rate check). If the error rate is too high, they conclude an eavesdropper is present and discard the key.

So, QKD is secure because it:

- generates a shared key using quantum properties, and
- detects interception through the increase of errors caused by measurement.

Example:

Think of QKD like sending a “sealed envelope” where opening it always leaves a mark. If someone tries to read the key while it’s being sent, Alice and Bob detect unusual errors and discard that key.

2. How can a modern web connection be made quantum-resistant by combining post-quantum cryptography with quantum key distribution while still working over normal internet infrastructure? (5 marks)

Answer:

A modern web connection (like HTTPS/TLS) can be made quantum-resistant by using a combined approach:

- **PQC in TLS handshake:** Replace RSA/ECC with PQC to make key exchange safe against quantum attacks.
- **Fast data encryption:** Use AES (or similar) to encrypt website data after the handshake.
- **Secure KMS:** Store, control, rotate, expire, and log keys to prevent leakage or misuse.
- **QKD where available:** Use quantum links to generate shared keys and detect interception attempts.
- **Hybrid + fallback:** Combine PQC + QKD when possible; otherwise use PQC-only TLS on normal internet.

Example :

Think of a bank website:

- The bank upgrades its website security to PQC.
- The bank’s head office data center also has a special secure fiber link using QKD to generate extra-secure keys.

- A key management system controls which server gets which key and when it expires.
- If QKD is unavailable for a user on normal internet, the connection still stays safe using PQC-only - so everything still works like normal HTTPS.

3. What is a Quantum Random Security Generator (QRSG) and why is it important? (4 Marks)

Answer:

A Quantum Random Security Generator (QRSG) is an advanced random number generation system that uses the principles of quantum physics, such as superposition and quantum uncertainty, to produce truly random numbers. In quantum systems, outcomes are naturally unpredictable due to the laws of physics. QRSG takes advantage of this natural randomness to generate secure cryptographic keys.

Unlike traditional Random Number Generators (RNGs), which depend on mathematical algorithms and may produce predictable patterns, QRSG generates numbers that are completely unpredictable and cannot be reproduced. This makes it much more secure for cryptographic applications.

QRSG is important because:

- 1) It provides higher entropy, meaning stronger and more reliable randomness.
- 2) It enhances the security of encryption keys and digital credentials.
- 3) It offers strong protection against brute-force, prediction, and even quantum attacks.
- 4) It improves the overall cybersecurity of Financial Supply Chains (FSCs) by ensuring secure transactions and data protection.

Therefore, QRSG plays a crucial role in strengthening modern cybersecurity systems, especially in the financial sector. QRSG :

4. What is the role of cryptography in Financial Supply Chains (FSCs)? (4 Marks)

Answer:

Cryptography plays a very important role in securing Financial Supply Chains (FSCs). Financial supply chains include online banking systems, digital payment platforms, fund transfers, and other financial transactions that operate through digital networks. Since these systems handle sensitive customer data and large amounts of money, strong security is necessary.

Cryptography protects these systems by converting sensitive information into encrypted form, so that only authorized users can read or access it. It ensures confidentiality, meaning private data remains secret, and integrity, meaning the data cannot be changed or tampered with during transmission. It also helps in authentication, verifying the identity of users and systems involved in transactions.

By preventing unauthorized access and securing communication between financial institutions, cryptography reduces the risk of cyber-attacks, fraud, and data leaks. Therefore, strong cryptographic techniques are essential for maintaining trust, safety, and smooth operation in financial supply chains.

5. Why is Elliptic Curve Cryptography (ECC) better than RSA for Wireless Sensor Networks (WSN)? (2 marks)

Answer:

Elliptic Curve Cryptography (ECC) is a modern security method used to protect data in networks.

- Uses smaller key size but gives strong security.
- Consumes less power, which is important for sensor nodes.
- Takes less time for encryption and decryption.
- Reduces network congestion.
- Improves overall network performance.

ECC is better for Wireless Sensor Networks because it is secure, fast, and energy efficient.

6. What are the main steps of the EAECC authentication process? (5 marks)

Answer:

EAECC is an improved security method that combines ECC and digital signature authentication.

Main Steps:

- Generate a private and public key.
- Create a digital signature for the message.
- Send the message with the signature.
- Verify the signature at the receiver side.

Advantages:

- Provides strong authentication.
- Protects against attacks.
- Reduces delay in communication.
- Saves energy in the network.

EAECC ensures secure and efficient communication in Wireless Sensor Networks.

7. What are the major security threats in 5G networks? (5 Marks)**Answer:**

5G networks provide high speed and low delay communication, but they also face many security risks.

- **Man-in-the-Middle (MitM) Attack:** Attackers intercept and modify communication between users.
- **Denial-of-Service (DoS) Attack:** Attackers overload the network and cause service disruption.
- **Authentication Bypass:** Weak authentication allows unauthorized access.
- **Rogue Base Stations:** Fake base stations can steal user data.
- **Quantum Threats:** Future quantum computers may break current encryption methods.

Although 5G offers advanced features, strong security mechanisms are necessary to protect data and services.

8. Explain the enhanced security mechanisms proposed for 5G networks. (5 Marks)**Answer:**

To improve 5G security, advanced protection methods are proposed.

- **AI-Based Intrusion Detection System (AI-IDS):** Detects attacks quickly and improves threat detection accuracy.

- **Blockchain Security:** Protects data integrity using decentralized trust.
- **Quantum Cryptography:** Provides very strong encryption against future quantum attacks.
- **Zero-Trust Architecture:** Ensures continuous authentication and strict access control.
- **Multi-layer Security Approach:** Combines different technologies for better protection.

Using AI, blockchain, quantum cryptography, and Zero-Trust policies can significantly improve 5G network security.

9. What is Generative AI (GAI) in network security? (4 Marks)

Answer:

Generative AI (GAI) is an advanced AI technology used to improve network security monitoring and warning systems.

- It can detect malicious traffic and cyber attacks.
- It handles multiple security tasks using one model.
- It reduces false alarms.
- It improves threat detection speed and accuracy.

GAI makes network security systems more intelligent and efficient.

10. What is a Multi-Agent based Network Security Monitoring System? (4 Marks)

Answer:

A Multi-Agent system is a security system where different AI agents work together to monitor and protect the network.

- Different agents perform different tasks.
- Planning agent manages the overall task.
- Memory agent stores security information.
- Review agent checks and verifies the results.

Multi-Agent systems improve flexibility, accuracy, and efficiency in network security monitoring.